

# Primary sources & further reading

Part 03 — Natural Language Processing · 03-nlp/papers/PAPERS.md · 03-nlp/resources/RESOURCES.md



## Papers — NLP & Transformers

`bash ../_tools/fetch_papers.sh 03-nlp` downloads these into this folder.

**Read #1 twice. Then read it again in a month.**

| # | Paper                                                                                          | Year | Why                                                                                                       | Link                             |
|---|------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------|----------------------------------|
| 1 | <b>Attention Is All You Need</b> — Vaswani et al.                                              | 2017 | The transformer. Eight pages that redirected the field. Read alongside <i>The Annotated Transformer</i> . | <a href="#">arXiv:1706.03762</a> |
| 2 | <b>BERT</b> — Devlin et al.                                                                    | 2018 | Bidirectional pretraining + fine-tuning; created the "pretrain then adapt" era                            | <a href="#">arXiv:1810.04805</a> |
| 3 | <b>Improving Language Understanding by Generative Pre-Training (GPT-1)</b> — Radford et al.    | 2018 | The decoder-only bet                                                                                      | <a href="#">PDF</a>              |
| 4 | <b>Neural Machine Translation by Jointly Learning to Align and Translate</b> — Bahdanau et al. | 2014 | Where attention was actually invented                                                                     | <a href="#">arXiv:1409.0473</a>  |

## Tokenization

| Paper                                                                                     | Takeaway                                          | Link                             |
|-------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------|
| Neural Machine Translation of Rare Words with Subword Units (BPE) — Sennrich et al., 2015 | BPE for NLP                                       | <a href="#">arXiv:1508.07909</a> |
| SentencePiece — Kudo & Richardson, 2018                                                   | Language-agnostic, no whitespace pre-tokenization | <a href="#">arXiv:1808.06226</a> |
| Subword Regularization — Kudo, 2018                                                       | The unigram LM tokenizer                          | <a href="#">arXiv:1804.10959</a> |

## Architecture & efficiency

| Paper                                                                                | Takeaway                                               | Link                             |
|--------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------|
| RoFormer (RoPE) — Su et al., 2021                                                    | Rotary position embeddings; in almost every modern LLM | <a href="#">arXiv:2104.09864</a> |
| Train Short, Test Long (ALiBi) — Press et al., 2021                                  | Length extrapolation via attention bias                | <a href="#">arXiv:2108.12409</a> |
| Fast Transformer Decoding: One Write-Head (MQA) — Shazeer, 2019                      | The KV-cache problem, stated early                     | <a href="#">arXiv:1911.02150</a> |
| GQA: Training Generalized Multi-Query Transformer Checkpoints — Ainslie et al., 2023 | The modern default                                     | <a href="#">arXiv:2305.13245</a> |
| FlashAttention — Dao et al., 2022                                                    | IO-aware exact attention                               | <a href="#">arXiv:2205.14135</a> |
| FlashAttention-2 — Dao, 2023                                                         | Better parallelism and work partitioning               | <a href="#">arXiv:2307.08691</a> |
| GLU Variants Improve Transformer (SwiGLU) — Shazeer, 2020                            | The modern FFN                                         | <a href="#">arXiv:2002.05202</a> |
| Root Mean Square Layer Normalization (RMSNorm) — Zhang & Sennrich, 2019              | Cheaper norm, same result                              | <a href="#">arXiv:1910.07467</a> |
| On Layer Normalization in the Transformer Architecture — Xiong et al., 2020          | Why Pre-LN beat Post-LN                                | <a href="#">arXiv:2002.04745</a> |
| T5: Exploring the Limits of Transfer Learning — Raffel et al., 2019                  | Everything as text-to-text; a huge ablation study      | <a href="#">arXiv:1910.10683</a> |
| Efficient Transformers: A Survey — Tay et al., 2020                                  | Map of the efficiency landscape                        | <a href="#">arXiv:2009.06732</a> |
| Mamba: Linear-Time Sequence Modeling — Gu & Dao, 2023                                | The serious non-attention challenger                   | <a href="#">arXiv:2312.00752</a> |

## Embeddings & retrieval (bridges to Module 08)

| Paper                                                                                  | Takeaway                                               | Link                             |
|----------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------|
| Efficient Estimation of Word Representations (word2vec) — Mikolov et al., 2013         | Meaning as geometry                                    | <a href="#">arXiv:1301.3781</a>  |
| Sentence-BERT — Reimers & Gurevych, 2019                                               | Usable sentence embeddings via siamese training        | <a href="#">arXiv:1908.10084</a> |
| Dense Passage Retrieval (DPR) — Karpukhin et al., 2020                                 | Dense retrieval beats BM25                             | <a href="#">arXiv:2004.04906</a> |
| Text Embeddings by Weakly-Supervised Contrastive Pre-training (E5) — Wang et al., 2022 | Modern embedding recipe; note the instruction prefixes | <a href="#">arXiv:2212.03533</a> |

## The best explainers ever written about transformers

- **The Illustrated Transformer** — Jay Alammar: <https://jalammar.github.io/illustrated-transformer/>

- **The Annotated Transformer** — Harvard NLP (paper + line-by-line code): <https://nlp.seas.harvard.edu/annotated-transformer/>
- **Let's build GPT from scratch** — Karpathy (2h video, builds it live): <https://www.youtube.com/watch?v=kCc8FmEb1nY>
- **3Blue1Brown — Attention in transformers, visually**: <https://www.3blue1brown.com/topics/neural-networks>
- **Transformer Circuits / induction heads** — Anthropic: <https://transformer-circuits.pub/>

## Resources — NLP & Transformers

### Start here

| Resource                                                               | Why                                                                                  | Link                                                                                                                  |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Karpathy — Let's build GPT from scratch</b>                         | 2 hours, builds a working transformer live. The best single artifact in this module. | <a href="https://www.youtube.com/watch?v=kCc8FmEb1nY">https://www.youtube.com/watch?v=kCc8FmEb1nY</a>                 |
| <b>The Illustrated Transformer</b> — Jay Alammar                       | The diagrams everyone else copies                                                    | <a href="https://jalammar.github.io/illustrated-transformer/">https://jalammar.github.io/illustrated-transformer/</a> |
| <b>The Annotated Transformer</b> — Harvard NLP                         | The paper with runnable code beside each paragraph                                   | <a href="https://nlp.seas.harvard.edu/annotated-transformer/">https://nlp.seas.harvard.edu/annotated-transformer/</a> |
| <b>Hugging Face NLP Course</b> (free)                                  | Practical, current, hands-on                                                         | <a href="https://huggingface.co/learn/nlp-course">https://huggingface.co/learn/nlp-course</a>                         |
| <b>Stanford CS224N — NLP with Deep Learning</b>                        | The university treatment; lectures on YouTube                                        | <a href="https://web.stanford.edu/class/cs224n/">https://web.stanford.edu/class/cs224n/</a>                           |
| <b>Speech and Language Processing</b> — Jurafsky & Martin (free draft) | The field's textbook, updated yearly                                                 | <a href="https://web.stanford.edu/~jurafsky/slp3/">https://web.stanford.edu/~jurafsky/slp3/</a>                       |
| <b>3Blue1Brown — Transformers &amp; attention</b>                      | Visual intuition for the matrix operations                                           | <a href="https://www.3blue1brown.com/topics/neural-networks">https://www.3blue1brown.com/topics/neural-networks</a>   |

## Repositories

| Repo                                                                                                          | What's inside                                                   |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <a href="https://github.com/karpathy/nanoGPT">https://github.com/karpathy/nanoGPT</a>                         | ~300 lines of readable GPT. Train one on your laptop.           |
| <a href="https://github.com/karpathy/minGPT">https://github.com/karpathy/minGPT</a>                           | The even smaller pedagogical version                            |
| <a href="https://github.com/karpathy/minbpe">https://github.com/karpathy/minbpe</a>                           | BPE in a few hundred clean lines — compare with your own        |
| <a href="https://github.com/huggingface/transformers">https://github.com/huggingface/transformers</a>         | The library; read <code>modeling_llama.py</code> once, properly |
| <a href="https://github.com/huggingface/tokenizers">https://github.com/huggingface/tokenizers</a>             | Fast Rust tokenizers with a Python API                          |
| <a href="https://github.com/openai/tiktoken">https://github.com/openai/tiktoken</a>                           | OpenAI's BPE tokenizer                                          |
| <a href="https://github.com/google/sentencepiece">https://github.com/google/sentencepiece</a>                 | The tokenizer behind T5, Llama and many others                  |
| <a href="https://github.com/Dao-AI-Lab/flash-attention">https://github.com/Dao-AI-Lab/flash-attention</a>     | The reference implementation                                    |
| <a href="https://github.com/UKPLab/sentence-transformers">https://github.com/UKPLab/sentence-transformers</a> | Sentence embeddings — the on-ramp to Module 08                  |
| <a href="https://github.com/explosion/spaCy">https://github.com/explosion/spaCy</a>                           | Industrial classical NLP: NER, parsing, pipelines               |
| <a href="https://github.com/bentrevett/pytorch-seq2seq">https://github.com/bentrevett/pytorch-seq2seq</a>     | Clear tutorials from RNN seq2seq up to transformers             |

Clone the starter set with: `bash ../_tools/clone_repos.sh 03-nlp`

## Interactive / visual tools

- **Transformer Explainer** (interactive GPT-2 in your browser) — <https://poloclub.github.io/transformer-explainer/>
- **BertViz** (attention visualisation) — <https://github.com/jessevig/bertviz>
- **Tiktokenizer** (see tokens live) — <https://tiktokenizer.vercel.app/>
- **Hugging Face tokenizer playground** — <https://huggingface.co/spaces/Xenova/the-tokenizer-playground>
- **Transformer Circuits** (mechanistic interpretability) — <https://transformer-circuits.pub/>

## Datasets & leaderboards

- Hugging Face Datasets — <https://huggingface.co/datasets>
- GLUE / SuperGLUE — <https://gluebenchmark.com/> · <https://super.gluebenchmark.com/>
- MTEB (embedding leaderboard — read this before choosing an embedding model) — <https://huggingface.co/spaces/mteb/leaderboard>
- Papers with Code — NLP — <https://paperswithcode.com/area/natural-language-processing>